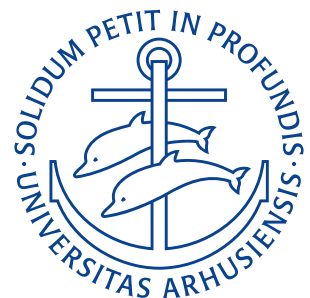


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Previous Exam Sets

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Lecture 21: Exam 2021

15. December 2025

Problem 21.1

Are the following functions $f(x)$, $-\infty < x < \infty$, even functions or odd functions? Give arguments.

- a) $f(x) = (\sin(x^2) + x^2 \cos(x)) \sin(x)$
- b) $f(x) = \cos(g(x))$, where $g(x)$ is an odd function
- c) $f(x) = \int_{-x}^x g(s) ds$, where the function f is such that the integral exists for $-\infty < x < \infty$

Derivation

- a) To determine if the given function is even or odd we see what happens when replacing the argument of the function with $-x$. We get

$$\begin{aligned} f(x) &= (\sin(x^2) + x^2 \cos(x)) \sin(x) \\ f(-x) &= (\sin((-x)^2) + (-x)^2 \cos(-x)) \sin(-x) \\ &= (\sin x^2 + x^2 \cos(x)) (-\sin(x)) \\ &= -f(x). \end{aligned}$$

Hence, the function is odd.

- b) We follow the same procedure and get

$$\begin{aligned} f(x) &= \cos(g(x)) \\ f(-x) &= \cos(g(-x)) \\ &= \cos(-g(x)) \\ &= \cos(g(x)) \\ &= f(x). \end{aligned}$$

Hence, the function is even.

- c) Again,

$$\begin{aligned} f(x) &= \int_{-x}^x g(s) ds \\ f(-x) &= \int_x^{-x} g(s) ds \\ &= - \int_{-x}^x g(s) ds \\ &= -f(x). \end{aligned}$$

Hence, the function is odd.

- a) Odd
- b) Even
- c) Odd.

RESULT

Problem 21.2

Find a period and the corresponding Fourier coefficients of the Fourier series of

- a) $f(x) = \sin\left(\frac{\pi}{5}\right) \sin\left(\frac{2x}{3}\right) + \cos\left(\frac{\pi}{5}\right) \sin\left(\frac{2x}{11}\right), -\infty < x < \infty$
 b) $f(x) = \frac{1}{2} \sin(\sqrt{8}x) + 3 \cos\left(\frac{x}{\sqrt{2}}\right) - 4, -\infty < x < \infty.$

Derivation

The general form of a Fourier series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right).$$

- a) We rewrite our given function as

$$f(x) = \underbrace{\sin\left(\frac{\pi}{5}\right)}_{b_{22}} \underbrace{\sin\left(\frac{22\pi x}{33\pi}\right)}_L + \underbrace{\cos\left(\frac{\pi}{5}\right)}_{b_6} \underbrace{\sin\left(\frac{6\pi x}{33\pi}\right)}_L.$$

Our period is therefore $p = 2L = 66\pi$ and our Fourier coefficients are $b_6 = \cos(\pi/5)$ and $b_{22} = \sin(\pi/5)$. All other Fourier coefficients are zero

- b) We rewrite as

$$f(x) = \underbrace{-4}_{a_0} + \underbrace{+3}_{a_1} \underbrace{\cos\left(\frac{\pi x}{\sqrt{2}\pi}\right)}_L + \underbrace{\frac{1}{2}}_{b_4} \underbrace{\sin\left(\frac{4\pi x}{\sqrt{2}\pi}\right)}_L.$$

Hence, $p = 2L = 2\sqrt{2}\pi$, $a_0 = -4$, $a_1 = 3$, $b_4 = \frac{1}{2}$ and all other Fourier coefficients are zero.

a) $p = 66\pi, \quad b_6 = \cos \frac{\pi}{5} \quad b_{22} = \sin \frac{\pi}{5}$ b) $p = 2\sqrt{2}\pi, \quad a_0 = -4, \quad a_1 = 3, \quad b_4 = \frac{1}{2}.$

RESULT

Problem 21.3

Find the Fourier cosine transform of

$$f(x) = \begin{cases} |x|, & -1 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Derivation

The Fourier cosine transform is given as

$$\hat{f}_C(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos(\omega x) \, dx.$$

Plugging in our function, we obtain

$$\begin{aligned} \hat{f}_C(\omega) &= \sqrt{\frac{2}{\pi}} \int_0^1 x \cos(\omega x) \, dx \\ &= \sqrt{\frac{2}{\pi}} \left[\frac{\cos(\omega x)}{\omega^2} + \frac{x \sin(\omega x)}{\omega} \right]_{x=0}^{x=1} \\ &= \sqrt{\frac{2}{\pi}} \left(\frac{\cos \omega}{\omega^2} - \frac{1}{\omega^2} + \frac{\sin \omega}{\omega} \right) \\ &= \sqrt{\frac{2}{\pi}} \frac{\omega \sin \omega + \cos \omega - 1}{\omega^2}. \end{aligned}$$

$$\hat{f}_C(\omega) = \sqrt{\frac{2}{\pi}} \frac{\omega \sin \omega + \cos \omega - 1}{\omega^2}$$

RESULT

Problem 21.4

Consider the PDE

$$u_t(x, t) + u_{xx}(x, t) = 0$$

where $0 \leq x \leq 1, 0 \leq t$.

- a) Find the general solution that fulfils the conditions

$$u(0, t) = u(1, t) = 0$$

for all t .

- b) Find the unique solution that fulfils the conditions

$$u(0, t) = u(1, t) = 0$$

for all t and

$$u_t(x, 0) = \sin(3\pi x) + 4\sin(2\pi x)$$

for $0 \leq x \leq 1$.

Derivation

- a) Using the method of separation of variables, we assume $u(x, t) = F(x)G(t)$ and get

$$F(x)G'(t) + F''(x)G(t) = 0$$

$$\frac{F''(x)}{F(x)} = -\frac{G'(t)}{G(t)} = k.$$

Which corresponds to the two ODEs

$$F''(x) - kF(x) = 0 \quad G'(t) + kG(t) = 0.$$

Our boundary conditions gives us

$$u(0, t) = F(0)G(t) = 0$$

$$\implies F(0) = 0$$

$$u(1, t) = F(1)G(t) = 0$$

$$\implies F(1) = 0.$$

We quickly see, that assuming $x = 0 \implies F(x) = G(t) = 0$ which is clearly uninteresting.

If we assume $k > 0$ then

$$F(x) = Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x}$$

$$F(0) = 0 \implies A = -B$$

$$F(1) = 0 \implies 0 = Ae^{\sqrt{k}} + Be^{-\sqrt{k}} \implies A = 0 = B.$$

Which is also uninteresting

Lastly, we assume $k < 0$ and get

$$F(x) = A\cos(\sqrt{-k}x) + B\sin(\sqrt{-k}x)$$

$$F(0) = 0 \implies A = 0$$

$$F(1) = 0 \implies B \sin(\sqrt{-k}) = 0.$$

We set $\sqrt{-k} = p$ and get

$$F(1) = B \sin(p) = 0 \implies p = p_n = n\pi, n = 1, 2, \dots$$

We choose $B = 1$ and get

$$F_n(x) = \sin(n\pi x).$$

From the lecture notes we have that

$$G_n(t) = B_n e^{n^2 \pi^2 t}.$$

Hence,

$$\begin{aligned} u_n(x, t) &= F_n(x) G_n(t) \\ &= B_n e^{n^2 \pi^2 t} \sin(n\pi x) \\ u(x, t) &= \sum_{n=1}^{\infty} u_n(x, t) \\ u(x, t) &= \sum_{n=1}^{\infty} B_n e^{n^2 \pi^2 t} \sin(n\pi x). \end{aligned}$$

b) From above, we have the solution

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{n^2 \pi^2 t} \sin(n\pi x).$$

The initial condition gives

$$\begin{aligned} u_t(x, 0) &= \sum_{n=1}^{\infty} B_n n^2 \pi^2 \sin(n\pi x) \\ &= \sin(3\pi x) + 4 \sin(2\pi x) \\ \implies B_3 &= \frac{1}{9\pi^2}, B_2 = \frac{1}{\pi^2}. \end{aligned}$$

All other coefficients B are zero, hence

$$u(x, t) = \frac{1}{9\pi^2} e^{9\pi^2 t} \sin(3\pi x) + \frac{1}{\pi^2} e^{4\pi^2 t} \sin(2\pi x).$$

$\begin{aligned} \text{a) } u(x, t) &= \sum_{n=1}^{\infty} B_n e^{n^2 \pi^2 t} \sin(n\pi x) \\ \text{b) } u(x, t) &= \frac{1}{9\pi^2} e^{9\pi^2 t} \sin(3\pi x) + \frac{1}{\pi^2} e^{4\pi^2 t} \sin(2\pi x). \end{aligned}$

RESULT

Problem 21.5

Consider the change of variables $(x, y) \leftrightarrow (p, q)$ given by

$$x(p, q) = p + 1$$

$$y(p, q) = p - q$$

and

$$p(x, y) = x - 1$$

$$q(x, y) = x - y - 1.$$

Rewrite

$$u_x(x, y) + u_{yy}(x, y) = 0$$

in terms of the variables p and q .

Derivation

We define

$$\bar{u}(p, q) = \bar{u}(p(x, y), q(x, y)) = u(x, y) \implies \bar{u}_x(p, q) + \bar{u}_{yy}(p, q) = 0.$$

The first derivative with respect to x is

$$\begin{aligned} \bar{u}_x(p, q) &= \bar{u}_p(p, q)p_x(x, y) + \bar{u}_q(p, q)q_x(x, y) \\ &= \bar{u}_p(p, q) \cdot 1 + \bar{u}_q(p, q) \cdot 1 \\ &= \bar{u}_p(p, q) + \bar{u}_q(p, q). \end{aligned}$$

The first derivative with respect to y is

$$\begin{aligned} \bar{u}_y(p, q) &= \bar{u}_p(p, q)p_y(x, y) + \bar{u}_q(p, q)q_y(x, y) \\ &= \bar{u}_p(p, q) \cdot 0 + \bar{u}_q(p, q) \cdot (-1) \\ &= -\bar{u}_q(p, q). \end{aligned}$$

And the second derivative with respect to y is

$$\begin{aligned} \bar{u}_{yy}(p, q) &= (-\bar{u}_q(p, q))_y \\ &= -\bar{u}_{qp}(p, q)p_y(x, y) - \bar{u}_{qq}(p, q)q_y(x, y) \\ &= -\bar{u}_{qp}(p, q) \cdot 0 - \bar{u}_{qq}(p, q) \cdot (-1) \\ &= \bar{u}_{qq}(p, q). \end{aligned}$$

Inserting these into the original expression yields

$$\bar{u}_p(p, q) + \bar{u}_q(p, q) + \bar{u}_{qq}(p, q) = 0.$$

$$\bar{u}_p(p, q) + \bar{u}_q(p, q) + \bar{u}_{qq}(p, q) = 0$$

RESULT

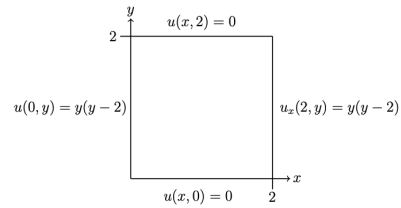
Problem 21.6

Consider the two-dimensional Poisson equation

$$u_{xx}(x, y) + u_{yy}(x, y) = xy(x-2)(y-2), 0 \leq x \leq 2, 0 \leq y \leq 2$$

subject to the mixed boundary conditions

$$\begin{aligned} u(0, y) &= y(y-2), & 0 \leq y \leq 2 \\ u_x(2, y) &= y(y-2), & 0 \leq y \leq 2 \\ u(x, 2) &= 0, & 0 \leq x \leq 2 \\ u(x, 0) &= 0, & 0 \leq x \leq 2. \end{aligned}$$



Derive the corresponding system of difference equations, using mesh size $h = 1$.

Derivation

The known boundary values are

$$\begin{aligned} u_{00} &= 0, & u_{10} &= 0, & u_{20} &= 0 \\ u_{01} &= -1, & u_{02} &= 0, \\ u_{12} &= 0, & u_{22} &= 0. \end{aligned}$$

The corresponding difference equation is

$$u_{i+1,j} + u_{i,j+1} + u_{i-1,j} + u_{i,j-1} - 4u_{i,j} = ijh^2(ih-2)(jh-2) = ij(i-2)(j-2).$$

To find meshpoint P_{11} we simply input the surrounding meshpoints into the above difference equation as

$$\begin{aligned} u_{21} + u_{12} + u_{01} + u_{10} - 4u_{11} &= 1^2 \cdot (-1) \cdot (-1) \\ u_{21} + 0 - 1 + 0 - 4u_{11} &= 1 \\ u_{21} - 4u_{11} &= 2. \end{aligned}$$

For the boundary meshpoint P_{21} we follow the same procedure and get

$$\begin{aligned} u_{31} + u_{22} + u_{11} + u_{20} - 4u_{21} &= 2 \cdot (0) \cdot (1) \\ u_{11} - 4u_{21} + u_{31} &= 0. \end{aligned}$$

Using the directional derivative, we find u_{31} as

$$\begin{aligned} u_x(2, 1) &= -1 \\ \Rightarrow \frac{u_{31} - u_{11}}{2} &= -1 \Rightarrow u_{31} = u_{11} - 2. \end{aligned}$$

Inserting this into the previous expression we get

$$2u_{11} - 4u_{21} = 2.$$

$$\begin{aligned} u_{21} - 4u_{11} &= 2 \\ 2u_{11} - 4u_{21} &= 2. \end{aligned}$$

RESULT

Problem 21.7

Consider the two-dimensional Laplace equation

$$u_{xx}(x, y) + u_{yy}(x, y) = 0, 0 \leq x \leq 2, 0 \leq y \leq 2$$

subject to the boundary conditions

$$u(0, y) = y(2 - y) - \frac{1}{2}, \quad 0 \leq y \leq 2$$

$$u(x, 2) = -\frac{1}{2}, \quad 0 \leq x \leq 2$$

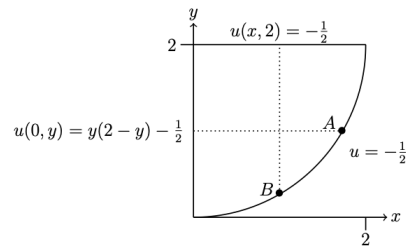
and $u(x, y) = -1/2$ for (x, y) on the curved line in the figure.

The curved line is a segment of a circle of radius 2 and centre

$(0, 2)$. The indicated points have coordinates $A = (\sqrt{3}, 1)$ and

$B = (1, 2 - \sqrt{3})$.

Find the approximate value of $u(1, 1)$, using mesh size $h = 1$.



Derivation

We use

$$\nabla^2 u(P_{11}) \approx \frac{2}{h^2} \left(\frac{1}{a(1+a)} u(A) + \frac{1}{b(1+b)} u(B) + \frac{1}{1+a} u(P) + \frac{1}{1+b} u(Q) - \frac{a+b}{ab} u(O) \right).$$

We have $h = 1$, $u(A) = u(B) = -1/2$: From the coordinates of A and B we see that $a = \sqrt{3} - 1$ and $b = \sqrt{3} - 1$ (this is the coefficient one has to multiply h with to move from the central point to A and B respectively (remember you are given coordinates and not relative distances)) Plugging this in we get

$$\begin{aligned} \nabla^2 u(P_{11}) &= \frac{2}{h^2} \left(\frac{1}{a(1+a)} u(A) + \frac{1}{b(1+b)} u(B) + \frac{1}{1+a} u(P) + \frac{1}{1+b} u(Q) - \frac{a+b}{ab} u(O) \right) = 0 \\ 0 &= \frac{1}{\sqrt{3}(\sqrt{3}-1)} \left(-\frac{1}{2} \right) + \frac{1}{\sqrt{3}(\sqrt{3}-1)} \left(-\frac{1}{2} \right) + \frac{1}{\sqrt{3}} \left(\frac{1}{2} \right) + \frac{1}{\sqrt{3}} \left(-\frac{1}{2} \right) - \frac{2}{\sqrt{3}-1} u(P_{11}) \\ \Rightarrow u(P_{11}) &= -\frac{1}{2\sqrt{3}}. \end{aligned}$$

$$u(P_{11}) = -\frac{1}{2\sqrt{3}}$$

RESULT

Lecture 23: Exam 2023

15. December 2025

Problem 23.1

Show that

$$f(x) = \begin{cases} 1 - x, & 0 < x < 1 \\ -1 - x, & -1 < x < 0 \end{cases}$$

is an odd function.

Derivation

To determine if the function is odd or even, we try inserting $-x$ as the argument to f . For $0 < x < 1$ we get:

$$f(-x) = 1 + x = -f(x).$$

And for $-1 < x < 0$ we get

$$f(-x) = -1 + x = -f(x).$$

Hence the function is odd.

Problem 23.2

Let $O_1(x)$ and $O_2(x)$ be odd functions and let $E(x)$ be an even function. Show that

$$f(x) = O_1(x) \left(e^{|O_2(x)E(x)|} + 1 \right)$$

is an odd function.

Derivation

We insert $-x$ into the function and get

$$\begin{aligned} f(-x) &= O_1(-x) \left(e^{|O_2(-x)E(-x)|} + 1 \right) \\ &= -O_1(x) \left(e^{|-O_2(x) \cdot E(x)|} + 1 \right) \\ &= - \left(O_1(x) \left(e^{|O_2(x)E(x)|} + 1 \right) \right) \\ &= -f(x). \end{aligned}$$

Hence, the function is odd.

Problem 23.3

Find a period and the corresponding Fourier coefficients of

a)

$$f(x) = \sin\left(\frac{\pi}{7}\right) + 2 \cos\left(\frac{2x}{\sqrt{2}\pi}\right) + 7 \sin\left(\frac{\sqrt{2}x}{3\pi}\right), \quad -\infty < x < \infty.$$

b)

$$f(x) = \sin(\sqrt{2}x) + \sin(3\sqrt{2}x), \quad -\infty < x < \infty.$$

Derivation

The general representation of a Fourier series is:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right).$$

a) We rewrite our given function as

$$f(x) = \underbrace{\sin\left(\frac{\pi}{7}\right)}_{a_0} + \underbrace{2}_{a_6} \cos\left(\underbrace{\frac{6\pi x}{3\sqrt{2}\pi^2}}_L\right) + \underbrace{7}_{b_2} \sin\left(\underbrace{\frac{2\pi x}{3\sqrt{2}\pi^2}}_L\right).$$

Hence, $p = 2L = 6\sqrt{2}\pi^2$, $a_0 = \sin(\pi/7)$, $a_6 = 2$, $b_2 = 7$ and all other Fourier coefficients are zero.

b) Once again, we rewrite as

$$f(x) = \underbrace{1}_{b_2} \cdot \sin\left(\underbrace{\frac{2\pi x}{\sqrt{2}\pi}}_L\right) + \underbrace{1}_{b_6} \sin\left(\underbrace{\frac{6\pi x}{\sqrt{2}\pi}}_L\right).$$

Hence, $p = 2L = 2\sqrt{2}\pi$, $b_2 = 1$, $b_6 = 1$ and all other Fourier coefficients are zero.

a) $p = 6\sqrt{2}\pi^2$, $a_0 = \sin\frac{\pi}{7}$, $a_6 = 2$, $b_2 = 7$ b) $p = 2\sqrt{2}\pi$, $b_2 = 1$, $b_6 = 1$.
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RESULT

Problem 23.4

Consider the function

$$f(x) = \sin(2x), 0 \leq x \leq \pi.$$

- Find the period and the corresponding Fourier coefficients of the expansion of $f(x)$.
- Find the period and the corresponding Fourier coefficients of the odd expansion of $f(x)$.

Derivation

- As $\sin$ is an odd function, the expansion of $f(x) = \sin(2x), 0 \leq x \leq \pi$ is simply $f_0(x) = \sin(2x), -\pi/2 \leq x \leq \pi/2$. Rewriting this we get

$$f_0(x) = \underbrace{1}_{b_1} \cdot \sin \underbrace{\frac{\pi x}{2}}_L.$$

Meaning $p = 2L = \pi$ and $b_1 = 1$ and all other Fourier coefficients are zero.

- The odd expansion of an odd function will also yield the same odd function, however, for the odd expansion the period will be twice as long, as $f_2(x) = \sin(2x), -\pi \leq x \leq \pi$. Rewriting, we get

$$f_2(x) = \underbrace{1}_{b_2} \cdot \sin \underbrace{\frac{2\pi x}{\pi}}_L.$$

Hence, $p = 2L = 2\pi$, $b_2 = 1$ and all other Fourier coefficients are zero.

a)	$p = \pi,$	$b_1 = 1$
b)	$p = 2\pi,$	$b_2 = 1.$

RESULT

Problem 23.5

Rewrite the PDE

$$u_{yy}(x, y) + u_x(x, y) = 0$$

in terms of the variables

$$p(x, y) = e^y, q(x, y) = e^x.$$

Derivation

We define

$$\bar{u}(p, q) = \bar{u}(p(x, y), q(x, y)) = u(x, y) \implies \bar{u}_{yy}(p, q) + \bar{u}_x(p, q) = 0.$$

The first derivative with respect to x is

$$\begin{aligned} \bar{u}_x(p, q) &= \bar{u}_p(p, q)p_x(x, y) + \bar{u}_q(p, q)q_x(x, y) \\ &= \bar{u}_p(p, q) \cdot 0 + \bar{u}_q(p, q) \cdot e^x \\ &= \bar{u}_q(p, q) \cdot q. \end{aligned}$$

The first derivative with respect to y is

$$\begin{aligned} \bar{u}_y(p, q) &= \bar{u}_p(p, q)p_y(x, y) + \bar{u}_q(p, q)q_y(x, y) \\ &= \bar{u}_p(p, q) \cdot e^y + \bar{u}_q(p, q) \cdot 0 \\ &= \bar{u}_p(p, q) \cdot p. \end{aligned}$$

The second derivative with respect to y is

$$\begin{aligned} \bar{u}_{yy}(p, q) &= \bar{u}_{pp}(p, q)p_y(x, y) \cdot e^y + \bar{u}_p(p, q)e^y \\ &= \bar{u}_{pp}(p, q)p^2 + \bar{u}_p(p, q)p. \end{aligned}$$

By insertion we get

$$\bar{u}_{pp}(q, p)p^2 + \bar{u}_p(q, p)p + \bar{u}_q(p, q)q = 0$$

RESULT

Problem 23.6

Consider the PDE

$$u_{xx}(x, t) + u_{tt}(x, t) = u(x, t), 0 \leq x \leq \infty, t \geq 0$$

and let $\hat{u}_C(\omega, t) = \mathcal{F}_C(u(x, t))$ be the Fourier cosine transform of $u(x, t)$ with respect to x . Derive an ODE for $\hat{u}_C(\omega, t)$ with respect to t . You may assume that

$$\mathcal{F}_C(u_{tt}(x, t)) = \frac{\partial^2}{\partial t^2} \mathcal{F}_C(u(x, t))$$

and $u_x(0, t) = 0$ for all t .

Derivation

We know that $\mathcal{F}_C(f'') = -\omega^2 \mathcal{F}_C(f) - \sqrt{2/\pi} f'(0)$. Hence, we can rewrite our expression as

$$\begin{aligned} -\omega^2 \hat{u}_C(\omega, t) - \sqrt{\frac{2}{\pi}} \cdot 0 + \frac{\partial^2}{\partial t^2} \hat{u}_C(\omega, t) &= \hat{u}_C(\omega, t) \\ -\omega^2 \hat{u}_C(\omega, t) + \frac{\partial^2}{\partial t^2} \hat{u}_C(\omega, t) &= \hat{u}_C(\omega, t). \end{aligned}$$

$$-\omega^2 \hat{u}_C(\omega, t) + \frac{\partial^2}{\partial t^2} \hat{u}_C(\omega, t) = \hat{u}_C(\omega, t)$$

RESULT

Problem 23.7

Consider the PDE

$$u_{yy}(x, y) + u_x(x, y) = 0, 0 \leq y \leq 1, x \geq 0$$

subject to the conditions

$$u(x, 0) = u(x, 1) = 0 \text{ for all } x$$

and

$$u(0, y) = 3 \sin(2\pi y) + 2 \sin(3\pi y) \text{ for } 0 \leq y \leq 1.$$

Find the solution $u(x, y)$.

Derivation

Using the method of separation of variables, we assume $u(x, y) = F(x)Q(y)$ and get

$$\begin{aligned} F(x)Q''(y) + F'(x)Q(y) &= 0 \\ \frac{Q''(y)}{Q(y)} &= -\frac{F'(x)}{F(x)} = k. \end{aligned}$$

Which leaves us with the two ODEs

$$\begin{aligned} Q''(y) - kQ(y) &= 0 \\ F'(x) - kF(x) &= 0. \end{aligned}$$

Our boundary conditions gives us

$$\begin{aligned} u(x, 0) &= F(x)Q(0) = 0 \\ \implies Q(0) &= 0 \\ u(x, 1) &= F(x)Q(1) = 0 \\ \implies Q(1) &= 0. \end{aligned}$$

Assuming $k = 0 \implies F(x) = Q(y) = 0$ which is an uninteresting case.

If we assume $k > 0$ then

$$\begin{aligned} Q(y) &= Ae^{\sqrt{k}y} + Be^{-\sqrt{k}y} \\ Q(0) = 0 &\implies A = -B \\ Q(1) = 0 &\implies 0 = Ae^{\sqrt{k}} + Be^{-\sqrt{k}} \implies A = 0 = B. \end{aligned}$$

This is also uninteresting.

Hence, our only interesting case arises for $k < 0$, for which we get

$$\begin{aligned} Q(y) &= A \cos(\sqrt{-k}y) + B \sin(\sqrt{-k}y) \\ Q(0) = 0 &\implies A = 0 \\ Q(1) = 0 &\implies B \sin(\sqrt{-k}) = 0. \end{aligned}$$

We set $\sqrt{-k} = p$ and get

$$Q(1) = B \sin p = 0 \implies p = p_n = n\pi, n = 1, 2, \dots$$

We choose $B = 1$ and get

$$Q_n(y) = \sin(n\pi y).$$

From the Lecture notes we have that

$$F_n(x) = B_n e^{n^2 \pi^2 x}.$$

Hence,

$$\begin{aligned} u_n(x, y) &= F_n(x) Q_n(y) \\ &= B_n e^{n^2 \pi^2 x} \sin(n\pi y) \\ u(x, y) &= \sum_{n=1}^{\infty} u_n(x, y) \\ u(x, y) &= \sum_{n=1}^{\infty} B_n e^{n^2 \pi^2 x} \sin(n\pi y). \end{aligned}$$

We now compare this with our initial condition:

$$\begin{aligned} u(0, y) &= 3 \sin(2\pi y) + 2 \sin(3\pi y) \\ \implies B_2 &= 3 \quad B_3 = 2 \end{aligned}$$

and all other Fourier coefficients are zero.

Hence,

$$u(x, y) = 3e^{4\pi^2 x} \sin(2\pi y) + 2e^{9\pi^2 x} \sin(3\pi y)$$

RESULT

Problem 23.8

Consider the two-dimensional Laplace equation

$$u_{xx}(x, y) + u_{yy}(x, y) = 0, 0 \leq x \leq 2, 0 \leq y \leq 2$$

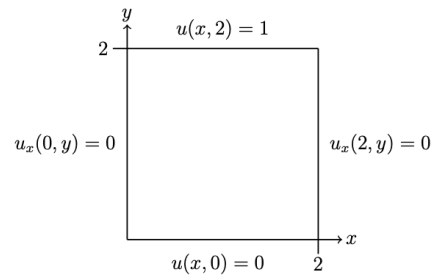
subject to the mixed boundary conditions

$$u_x(0, y) = 0, \quad 0 \leq y \leq 2$$

$$u_x(2, y) = 0, \quad 0 \leq y \leq 2$$

$$u(x, 2) = 1, \quad 0 \leq x \leq 2$$

$$u(x, 0) = 0, \quad 0 \leq x \leq 2.$$



Derive the corresponding system of difference equations, using mesh size $h = 1$.

Derivation

We have the known boundary values

$$u_{00} = 0,$$

$$u_{10} = 0,$$

$$u_{20} = 0,$$

$$u_{02} = 1,$$

$$u_{12} = 1,$$

$$u_{22} = 1.$$

And a corresponding difference equation of

$$u_{i+1,j} + u_{i,j+1} + u_{i-1,j} + u_{i,j-1} - 4u_{i,j} = 0.$$

We start by calculating the unknown meshpoint P_{01} using the difference equation as

$$u_{1,1} + u_{0,2} + u_{-1,1} + u_{0,0} - 4u_{0,1} = 0$$

$$u_{1,1} + 1 + u_{-1,1} + 0 - 4u_{0,1} = 0$$

$$u_{1,1} + u_{-1,1} - 4u_{0,1} = -1.$$

We find $u_{-1,1}$ using the directional derivative as

$$0 = \frac{u_{1,1} - u_{-1,1}}{2} \implies u_{-1,1} = u_{1,1}.$$

Hence,

$$2u_{1,1} - 4u_{0,1} = -1.$$

For the meshpoint P_{11} we get

$$u_{21} + u_{12} + u_{01} + u_{10} - 4u_{11} = 0$$

$$u_{21} + 1 + u_{01} + 0 - 4u_{11} = 0$$

$$u_{21} + u_{01} - 4u_{11} = -1.$$

For the meshpoint P_{21} we get

$$u_{31} + u_{22} + u_{11} + u_{20} - 4u_{21} = 0$$

$$u_{31} + 1 + u_{11} + 0 - 4u_{21} = 0$$

$$u_{31} + u_{11} - 4u_{21} = -1.$$

We find u_{31} using the directional derivative as

$$0 = \frac{u_{31} - u_{11}}{2} \implies u_{31} = u_{11}.$$

Hence

$$2u_{11} - 4u_{21} = -1.$$

$\begin{aligned} 2u_{11} - 4u_{01} &= -1 \\ u_{21} + u_{01} - 4u_{11} &= -1 \\ u_{31} + u_{11} - 4u_{21} &= -1. \end{aligned}$
--

RESULT

Lecture 22: Exam 2022

15. December 2025

Problem 22.1

Consider the function

$$f(x) = \begin{cases} 1, & -\infty < x < 0 \\ 0, & x = 0 \\ 1 + x, & 0 < x < \infty. \end{cases}.$$

Find an even function $f_1(x)$ and an odd function $f_2(x)$ such that

$$f(x) = f_1(x) + f_2(x), -\infty < x < \infty.$$

Problem 22.2

Find a period and the corresponding Fourier coefficients of the Fourier series of

- a) $f(x) = 2 \sin\left(\frac{x}{\pi}\right) + 3 \sin\left(\frac{2x}{3\pi}\right), -\infty < x < \infty$
 - b) $f(x) = 2 \sin(\sqrt{2}\pi x) + 3 \cos\left(\frac{2\pi x}{\sqrt{2}}\right) - \sin\left(\frac{\pi}{5}\right), -\infty < x < \infty.$
-

Problem 22.3

Find the Fourier transform of $f(x), -\infty < x < \infty$, defined as

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ -1, & -1 < x < 0 \\ 0, & \text{otherwise.} \end{cases}$$

Problem 22.4

Consider the PDE

$$u_{xx}(x, y, t) + u_y(x, y, t) = u_t(x, y, t).$$

Use the method of separation of variables to convert the above PDE into three ODEs.

Problem 22.5

Consider the change of variables $(x, y) \leftrightarrow (p, q)$ given by

$$\begin{aligned} x(p, q) &= p + 1 \\ y(p, q) &= p + q \end{aligned}$$

and

$$p(x, y) = x - 1$$

$$q(x, y) = y - x + 1.$$

Rewrite

$$u_{xy}(x, y) = 0$$

in terms of the variables p and q . You may assume that all functions are twice continuously differentiable.

Problem 22.6

Consider the two-dimensional Laplace equation

$$u_{xx}(x, y) + u_{yy}(x, y) = 0, 0 \leq x \leq 2, 0 \leq y \leq 2$$

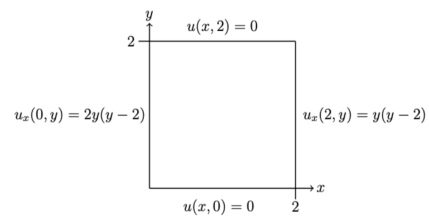
subject to the mixed boundary conditions

$$u_x(0, y) = 2y(y - 2), \quad 0 \leq y \leq 2$$

$$u_x(2, y) = y(y - 2), \quad 0 \leq y \leq 2$$

$$u(x, 2) = 0, \quad 0 \leq x \leq 2$$

$$u(x, 0) = 0, \quad 0 \leq x \leq 2.$$



Derive the corresponding system of difference equations, using mesh size $h = 1$.

Problem 22.7

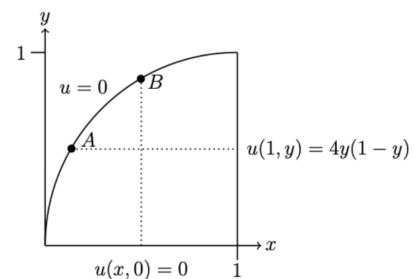
Consider the two-dimensional Laplace equation

$$u_{xx}(x, y) + u_{yy}(x, y) = 0$$

on the region illustrated in the figure and subject to the boundary conditions

$$u(1, y) = 4y(1 - y), \quad 0 \leq y \leq 1$$

$$u(x, 0) = 0, \quad 0 \leq x \leq 1$$



and $u(x, y) = 0$ for (x, y) on the curved line in the figure. The curved line is a segment of a circle with radius 1 and centre $(1, 0)$. The indicated points have coordinates $A = \left(1 - \frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ and $B = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

Find the approximate value of $u\left(\frac{1}{2}, \frac{1}{2}\right)$, using mesh size $h = \frac{1}{2}$.

Lecture 24: Exam 2024

15. December 2025

Problem 24.1

Show that

$$f(x) = \begin{cases} xe^{-|x|}, & 0 < x < 1 \\ -xe^{-|x|}, & -1 < x < 0 \end{cases}$$

is an even function.

Problem 24.2Let $O_1(x)$ and $O_2(x)$ be odd functions and let $E(x)$ be an even function. Show that

$$f(x) = \frac{O_1(x)E(x)}{O_2(x)} |O_1(x)|$$

is an even function.

Problem 24.3

Find a period and the corresponding Fourier coefficients of

a)

$$f(x) = \sin\left(\frac{\pi}{3}\right) + 7 \sin\left(\frac{3x}{\sqrt{3}\pi}\right) + 4 \cos\left(\frac{\sqrt{3}x}{4\pi}\right), -\infty < x < \infty.$$

b)

$$f(x) = 2 \sin\left(\frac{x}{\sqrt{2}}\right) + 4 \cos\left(\sqrt{2}x\right), -\infty < x < \infty.$$

Problem 24.4

Consider the function

$$f(x) = \cos(3x), 0 \leq x \leq \frac{2\pi}{3}.$$

- a) Find the period and the corresponding Fourier coefficients of the expansion of $f(x)$.
 - b) Find the period and the corresponding Fourier coefficients of the even expansion of $f(x)$.
-

Problem 24.5

Rewrite the PDE

$$u_{yy}(x, y) + u_x(x, y) = 0$$

in terms of the variables

$$p(x, y) = \frac{y}{2}, q(x, y) = e^x - 2.$$

Problem 24.6

Consider the PDE

$$u_{xxx}(x, t) + u_t(x, t) = u_x(x, t) + u(x, t), -\infty < x < \infty, t \geq 0$$

and let $\hat{u}(\omega, t) = \mathcal{F}(u(x, t))$ be the Fourier transform of $u(x, t)$ with respect to x . Derive an ODE for $\hat{u}(\omega, t)$ with respect to t . You may assume that the Fourier transforms of $u(x, t)$, $u_t(x, t)$, $u_x(x, t)$ and $u_{xxx}(x, t)$ exist and that

$$\mathcal{F}(u_t(x, t)) = \frac{\partial}{\partial t} \mathcal{F}(u(x, t)).$$

Problem 24.7

Consider the PDE

$$u_{yy}(x, y) + u_x(x, y) = 0, 0 \leq y \leq 1, x \geq 0$$

subject to the conditions

$$u_y(x, 0) = u_y(x, 1) = 0 \text{ for all } x$$

and

$$u(0, y) = 3 \cos(2\pi y) + 2 \cos(3\pi y) \text{ for } 0 \leq y \leq 1.$$

Find the solution $u(x, y)$.

Problem 24.8

Consider the PDE

$$2u_{xx}(x, y, t) - u_y(x, y, t) + u_t(x, y, t) = u(x, y, t).$$

Use the method of separation of variables to convert the above PDE into three ODEs.
